

Ben Pegg

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Education

University of Illinois Urbana-Champaign

Expected May 2028

Bachelor of Science in Computer Engineering

Relevant Coursework: Computer Organization & Design, VLSI Design, Digital Systems Lab, Data Structures & Algorithms

Skills

Engineering: Vivado, Vitis, Virtuosio, VCS, Verdi, Git, FPGAs (Spartan-7 series), Bench Test Equipment (Power Supply, Oscilloscope, Function Generator, Logic Analyzer), KiCad, MCUXpresso, STM32, NXP LPC

Protocols: CAN, I2C, SPI, UART, AXI4

Languages: SystemVerilog, C, C++, Python, Assembly

Experience

Illini Solar Car

Battery Management System Lead, Electrical Team Member

Sept 2024 – Present

- Redesigned the BMS PCB in KiCad after diagnosing critical hardware flaws in CAN, I2C, and power-path switching
- Led a team of 5 engineers to design fans firmware in C++ for an NXP microcontroller to implement PWM-controlled custom fan curves that optimize power consumption for the battery box and motor control controller cooling system
- Team finished first at the Formula Sun Grand Prix 2025 in the single-occupancy vehicle class

Winter Lodge

Front Desk Staff, Ice Guard

Sept 2021 – Jan 2025

- Provided efficient customer service in fast-paced environment and ensured a safe environment for hundreds of daily guests

Projects

FPGA IR Tracking and Parallax Graphics System

- Designed a real-time parallax graphics system on a Spartan-7 FPGA, utilizing IR light tracking to map the viewer's movement
- Wrote I2C drivers for an OV7670 camera to interface with the FPGA and optimize for light sensitivity and resolution
- Implemented a hardware-based centroid detection algorithm to calculate the IR light's coordinates and dynamically update the final multi-layer rendering on an HDMI display

RISC-V 5-Stage Pipelined Processor

- Designed a 5-stage pipelined processor using the RV32I ISA in SystemVerilog
- Implemented static branch prediction with pipeline flushing to resolve control hazards, minimize misprediction penalties
- Engineered hazard detection logic through full data forwarding to resolve data hazards without stalling
- Verified processor using a SystemVerilog constrained-random testbench, achieving 100% coverage of the RV32I instruction set

STM32 Macropad

- Designed PCB schematic and layout for a 9-key, dual rotary encoder macropad in KiCad using an STM32 microcontroller
- Developed C firmware for key matrix scanning, rotary encoder input, USB HID communication, and layer-switching
- Soldered and tested boards, validating and debugging board behavior using bench test equipment

CHIP-8 Emulator

- Created a CHIP-8 emulator in C++ by implementing the complete fetch-decode-execute cycle for all 35 original opcodes
- Built the emulator's core architecture, including virtual memory, registers, stack for subroutine calls, program/index counters
- Integrated SDL2 library to handle emulator input/output by rendering the display, mapping the keypad, and generating audio